

Consumption of vegetables alters morning urinary 6-sulfatoxymelatonin concentration.

Melatonin, which is contained in certain vegetables, may have an influence on circulatory melatonin concentrations. This study examined the effects of the consumption of vegetables on 6-sulfatoxymelatonin concentrations in morning urine. Ninety-four healthy women aged 24-55 were recruited through a city public health center in Japan. The women randomly allocated to the intervention group were requested to consume high amounts of six selected vegetables, with a target of 350 g/day for 65 days, while those in the control group were asked to avoid the same six vegetables during the same period. First-void morning urine was collected before and at the end of the intervention period, and creatinine-adjusted 6-sulfatoxymelatonin concentrations were measured. At the end of the intervention period, daily mean intake of melatonin from the six vegetables was 1288.0 ng in the intervention group and 5.3 ng in the control group. In the intervention group, the mean concentration of 6-sulfatoxymelatonin changed from 48.1 [95% confidence interval (CI): 40.4-57.2] ng/mg creatinine to 49.6 (95% CI: 42.8-57.3) ng/mg creatinine across the intervention period. In the control group, the mean concentration of 6-sulfatoxymelatonin changed from 55.5 (95% CI: 48.7-63.2) ng/mg creatinine to 50.8 (95% CI: 44.0-58.7) ng/mg creatinine across the intervention period. A comparison of the two groups with regard to the changes in the 6-sulfatoxymelatonin concentrations across the intervention period showed a significant difference ($P = 0.03$). The results indicate that increased consumption of vegetables raises circulatory melatonin concentrations.